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TECHNICAL SUMMARY

High-Speed PMSM Control Drives

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ABSTRACT

A Direct Torque Control drive for a fuel-cell compressor motor on a 200 kW, 450 V inverter, validated on hardware to 60,000 rpm. This project develops and implements a high-speed PMSM drive using Direct Torque Control. The motor starts open-loop and then transitions smoothly into closed-loop control. The drive pairs a 200 kW XM3 inverter on a 450 V DC bus with a fuel-cell compressor PMSM, and uses active cooling for thermal management. The work is related to an IEEE APEC 2025 paper comparing DTC on commercial and laboratory-developed high-speed PMSMs.

Validated on hardware	60,000 rpm
XM3 inverter	200 kW
DC bus	450 V
Control method	DTC

1. Problem and motivation

Fuel-cell systems use high-speed compressors to supply air, and the motors that drive them must start reliably and hold speed under changing load. At standstill there is no reliable flux or position information, so a high-speed DTC drive needs a safe way to start before closed-loop control can take over.

Because the drive is high-power and high-speed, heat and protection matter as much as the control law.

2. System description

- 200 kW XM3 inverter on a 450 V DC bus
- Fuel-cell compressor PMSM
- TI LaunchPad controller
- Active cooling for thermal management

3. Methodology

- Developed DTC with a seamless open-loop to closed-loop transition strategy
- Designed and tuned PI controllers for stable closed-loop operation under varying load
- Integrated the high-power inverter with the compressor motor
- Engineered active cooling to keep the system stable and prevent overheating at high speed

4. Figures

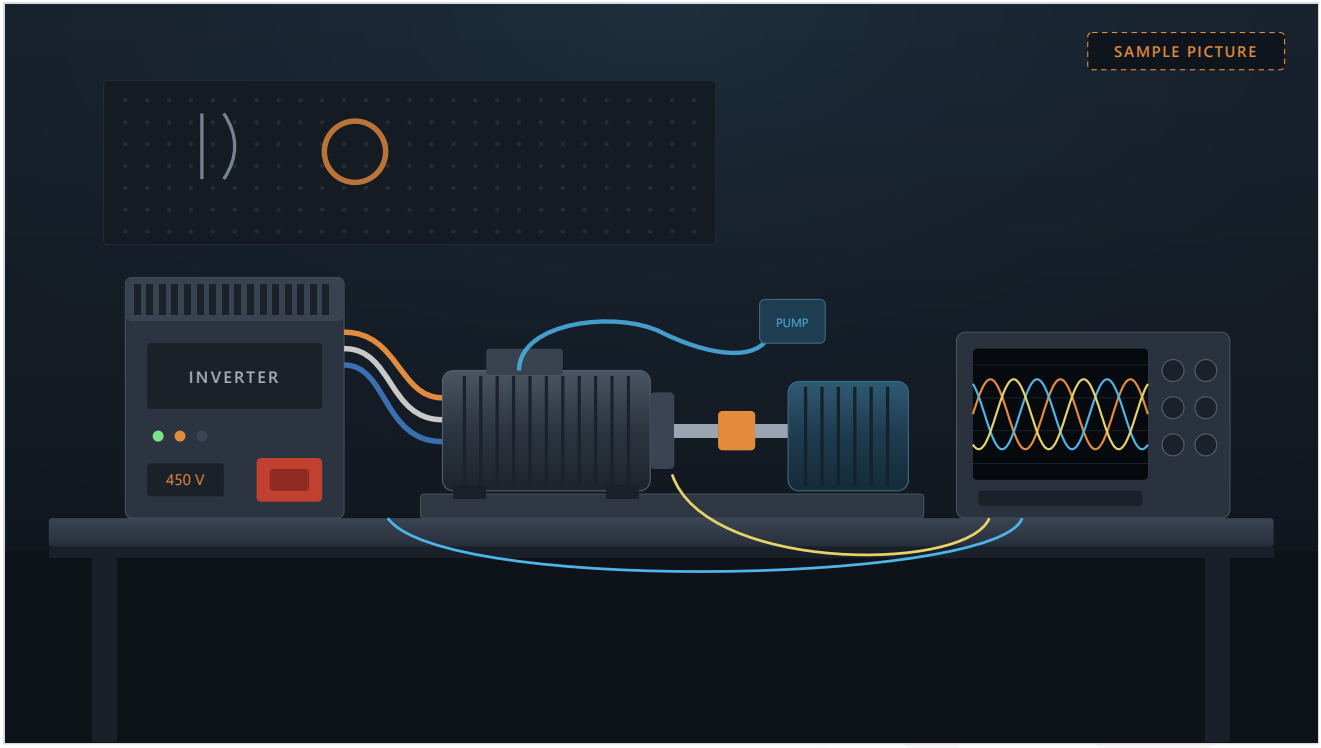


Figure 1. Motor test bench: inverter, high-speed PMSM coupled to a load, active cooling, and oscilloscope.

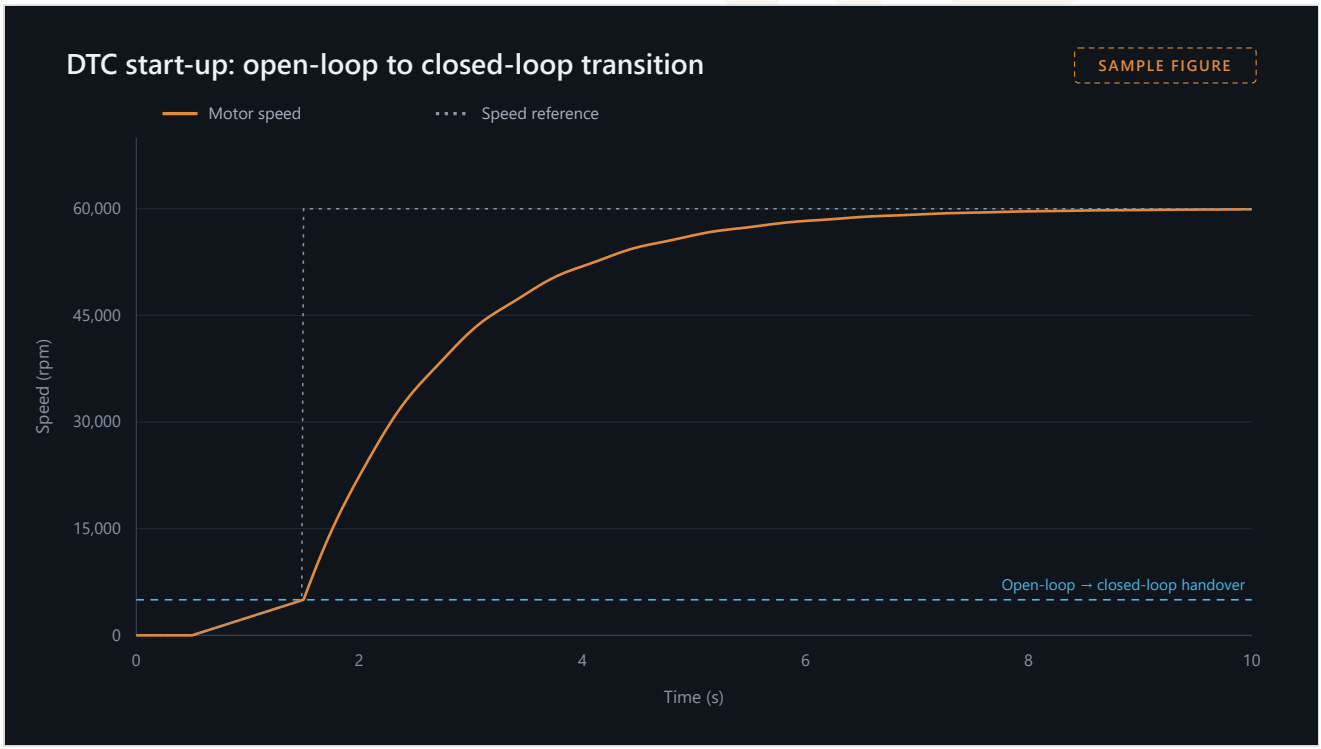


Figure 2. Start-up with the open-loop to closed-loop transition, up to 60,000 rpm.

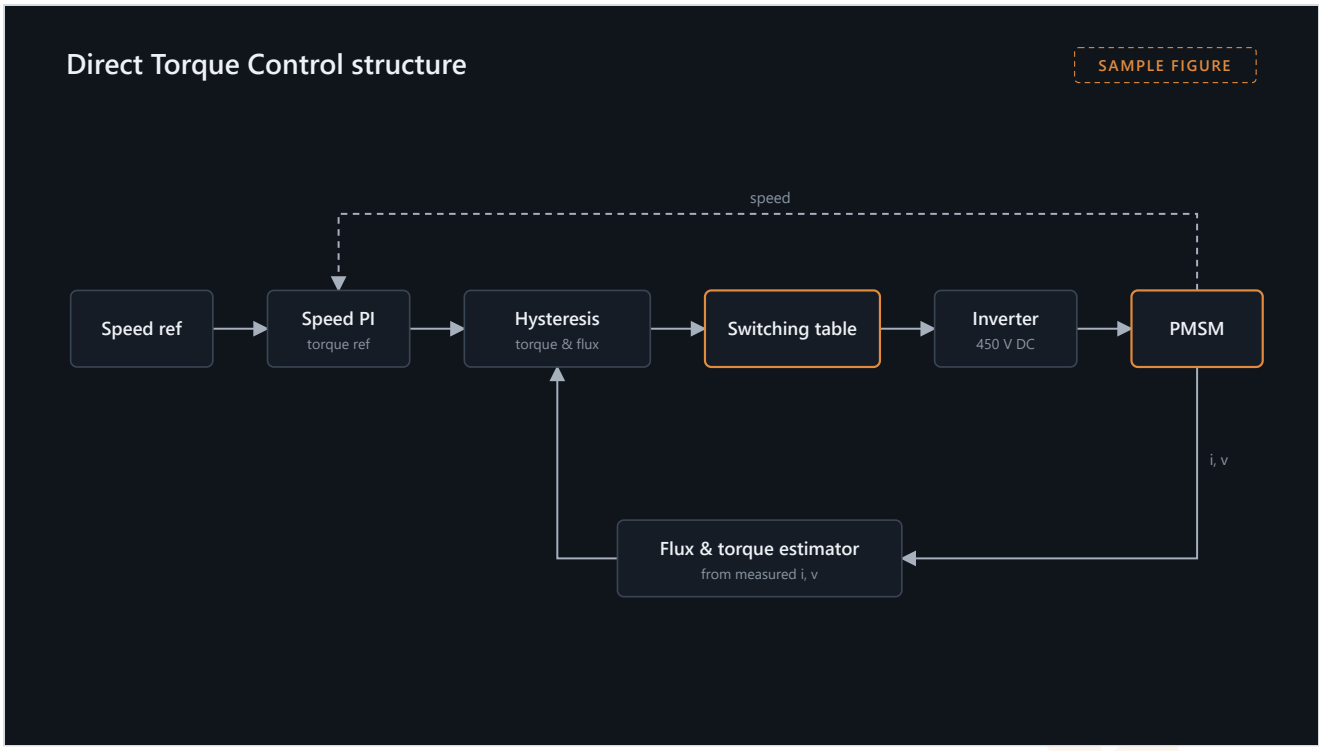


Figure 3. DTC structure: hysteresis torque and flux control with a switching table.

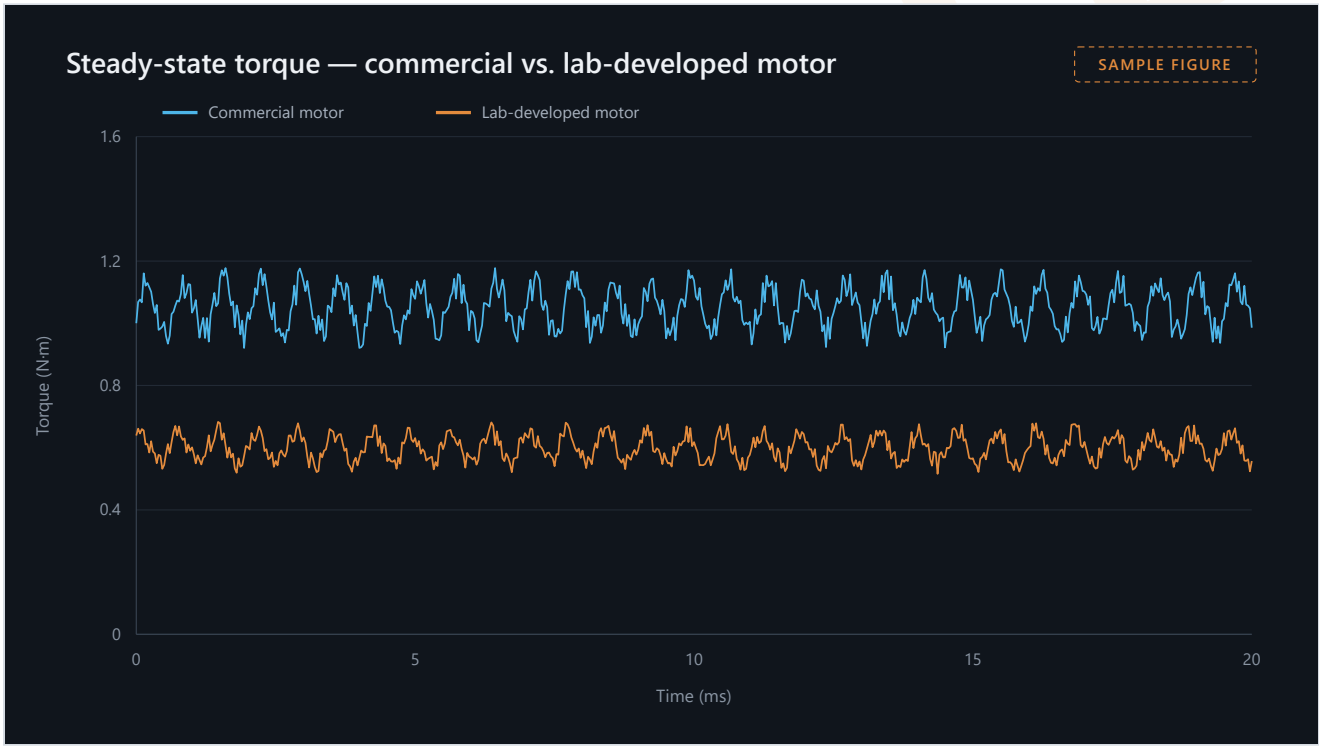


Figure 4. Steady-state torque of the commercial and lab-developed motors.

5. Results

- Validated on hardware up to 60,000 rpm
- Stable high-speed performance under varying load conditions
- Published comparison of DTC on commercial and lab-developed PMSMs at IEEE APEC 2025

6. Team and contribution

With M. Karimi-Ghartemani and S. Choi at Mississippi State University. APEC 2025 co-authors: M. Moniruzzaman, M. R. Rahman, M. K. Islam, S. Choi and M. Karimi-Ghartemani.

Contribution: Developed and implemented the DTC drive, including the open-loop to closed-loop transition, tuned the PI controllers, integrated the inverter with the motor, and set up the active cooling.

Tools: MATLAB/Simulink, PLECS, RT Box, TI LaunchPad, PI control

7. References

1. APEC 2025 paper (DOI): https://doi.org/10.1109/APEC48143.2025.10977544

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